Organization of Gene

“Genetic Code & Gene Expression”

SGL - 5

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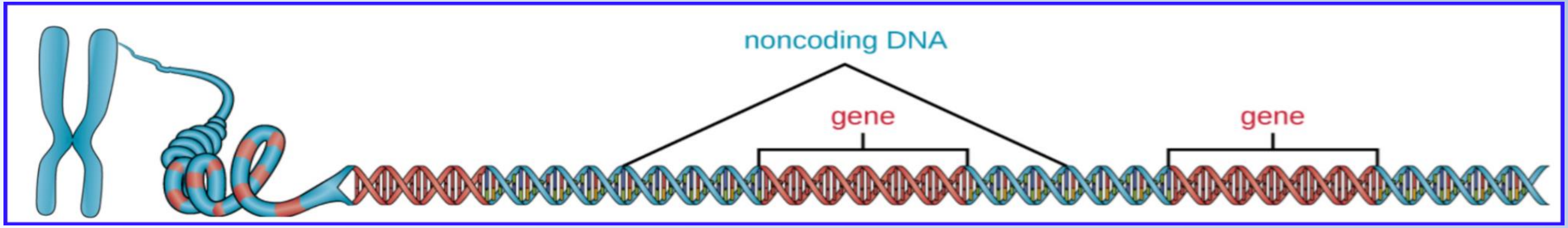
Molecular Biology :

Gene Characteristic:

- The term Gene refer to a piece of DNA containing the information for a single specific polypeptide “one gene, one polypeptide”.
- Now often use the term ‘gene’ as being synonymous with ‘open reading frame’ (ORF), i.e. the region between the start and stop codons.
- This definition more difficult, since the region of the chromosome that contains the information for a specific polypeptide may be many times longer than the actual coding sequence.
- The DNA contains many regions of noncoding DNA that do not encode proteins or stable RNA products and commonly found in areas prior to the start of coding of genes as well as in intergenic regions.
- The noncoding of DNA regions are nevertheless important for the viability of the cell; influence the phenotype in other ways and some region of DNA sequences are coding for rRNA and tRNA molecules.
- Any gene have Start Codon & Stop Codon



Molecular Biology : *Fundamentals*

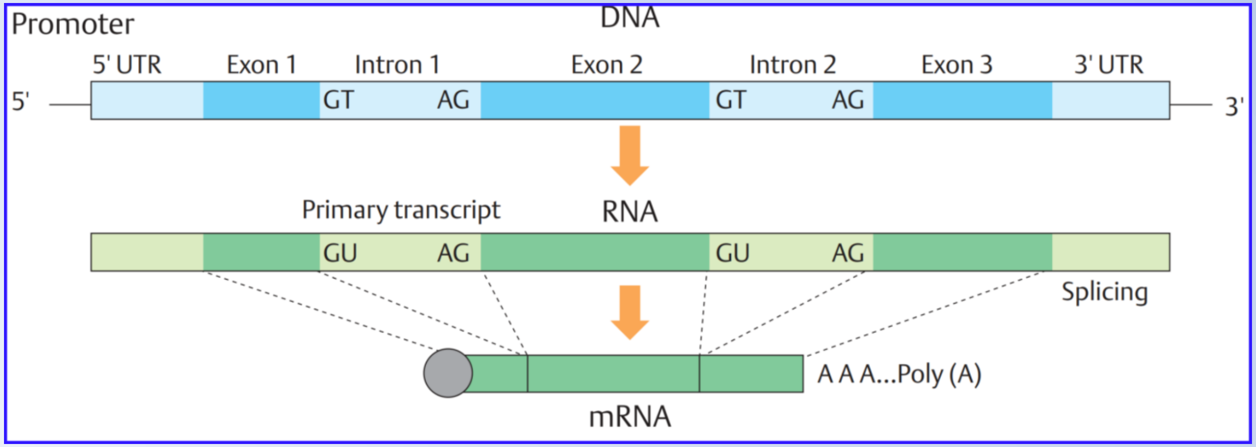
Gene: Exons and introns

- In bacteria there is generally a simple one-for-one relationship between the coding sequence of the DNA, the mRNA and the protein.

- In eukaryotic cells, the initial transcript is many times longer than that needed for translation into the final protein.

- It contains blocks of sequence (*introns*) which are removed by processing to generate the final mRNA for translation.

- One important consequence of the presence of introns is that several different variants of a protein can be produced from the same coding region of the genome, by differential processing of the mRNA.



***Molecular Biology :** Gene expression*

: Genetic Code

- Translation of the mRNA template converts nucleotide-based genetic information into the “language” of amino acids to create a protein product.

- A protein sequence consists of 20 commonly occurring amino acids. Each amino acid is defined within the mRNA by a triplet of nucleotides (triplet codon) called a (codon).

- The relationship between an mRNA codon and its corresponding amino acid is called the genetic code.

- The genetic code is the set of biological rules by which DNA nucleotide base pair sequences are translated into corresponding sequences of amino acids.

- Genes do not code for proteins directly, but do so through a messenger molecule (mRNA).

- The genetic code also includes sequences for the beginning (start codon) and for the end (stop codon) of the coding region.

- The genetic code is universal; the same codons are used by different organisms.

Molecular Biology : *Gene expression*

: Genetic Code

- The three-nucleotide code means that there is a total of 64 possible combinations, this number is greater than the number of amino acids and a given amino acid is encoded by more than one codon.

- Each codon corresponds to one amino acid, but one amino acid may be coded for by different codons (***Redundancy of the code***).

- For example, there are two possibilities to code for the amino acid phenylalanine: UUU and UUC, and there are six possibilities to code for the amino acid serine: UCU, UCC, UCA, UCG, AGU, and AGC.

second letter

first letter	U	C	A	G	third letter
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC	UCC	UAC	UGC	C
	UUA Leu	UCA	UAA stop	UGA stop	A
	UUG	UCG	UAG stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC	CCC	CAC	CGC	C
	CUA	CCA	CAA Gln	CGA	A
	CUG	CCG	CAG	CGG	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC	ACC	AAC	AGC	C
	AUA	ACA	AAA Lys	AGA Arg	A
	AUG Met	ACG	AAG	AGG	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC	GCC	GAC	GGC	C
	GUA	GCA	GAA Glu	GGA	A
	GUG	GCG	GAG	GGG	G

Molecular Biology : *Gene expression*

: Genetic Code

- The AUG, specifying the amino acid methionine, and act as the start codon.

- Typically, the first two positions in a codon are important for determining which amino acid will be incorporated into a growing polypeptide.

- The third position, called the ***(Wobble position)***, is less critical. In some cases, if the nucleotide in the third position is changed, the same amino acid is still incorporated.

second letter

first letter	U	C	A	G	third letter
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr **UAA stop** **UAG stop**	UGU Cys UGC Cys **UGA stop** UGG Trp	**U** **C** **A** **G**
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg	**U** **C** **A** **G**
A	AUU Ile AUC Ile AUA Ile **AUG Met**	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg	**U** **C** **A** **G**
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly	**U** **C** **A** **G**

Molecular Biology :

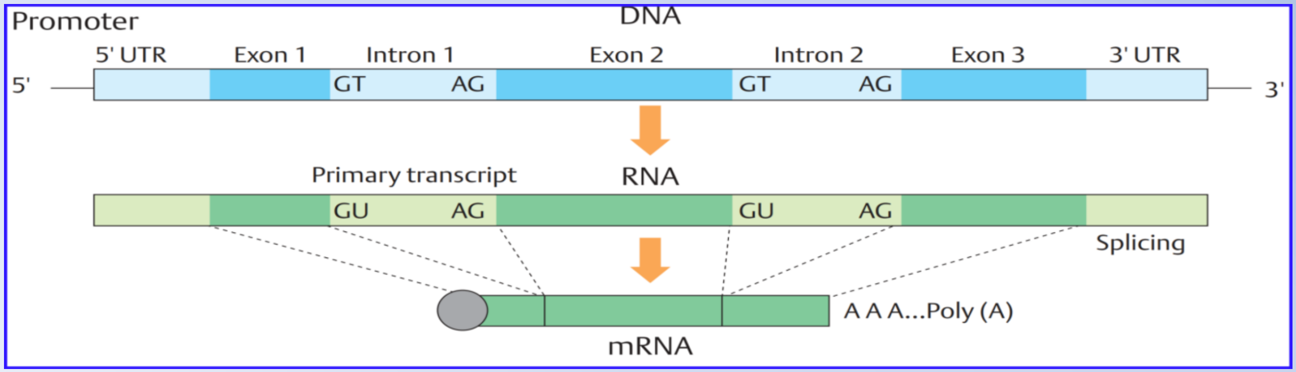
What Are Genes? :

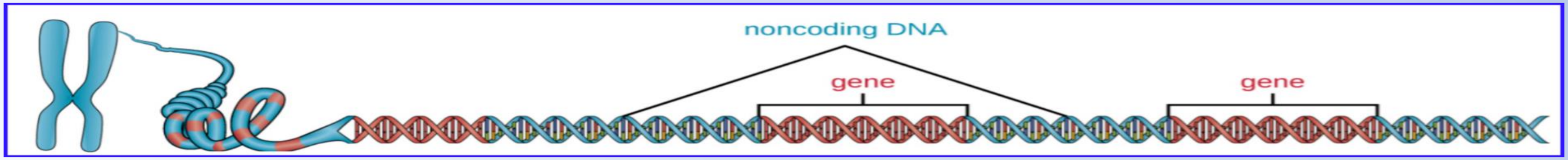
DNA serves two essential functions that deal with cellular information:

- DNA is the genetic material responsible for inheritance and is passed from parent to offspring for all life on earth. To preserve the integrity of this genetic information, DNA must be replicated with great accuracy, with minimal errors that introduce changes to the DNA sequence.

- The DNA is to direct and regulate the construction of the proteins necessary to a cell for growth and reproduction in a particular cellular environment.

The Gene contains blocks of sequences of *(Exons & Introns)*, the intros are removed by processing to generate the final mRNA for translation from





Molecular Biology : *Fundamentals*

Gene expression:

- DNA is the fundamental genetic material that carries information in living organism, The information contained in the nucleotide sequence of a gene must be converted into useful biological function and This accomplished by Proteins.

- Gene expression is the synthesis of a specific protein with a sequence of amino acids that is encoded in the gene

- Cells access the information stored in DNA by creating mRNA to direct the synthesis of proteins through the process of translation. And the three main types of RNA directly involved in protein synthesis.

- Neither of these types of RNA carries instructions to direct the synthesis of a polypeptide, but they play other important roles in protein synthesis.

- The mRNA serves as a photocopy of specific information needed at a particular point in time that serves as the instructions to make a protein, and the protein synthesis processes requires two major steps:

  a- Transcription   b- Translation

- The processes of transcription and translation are collectively referred to as gene expression.

Molecular Biology : Fundamentals

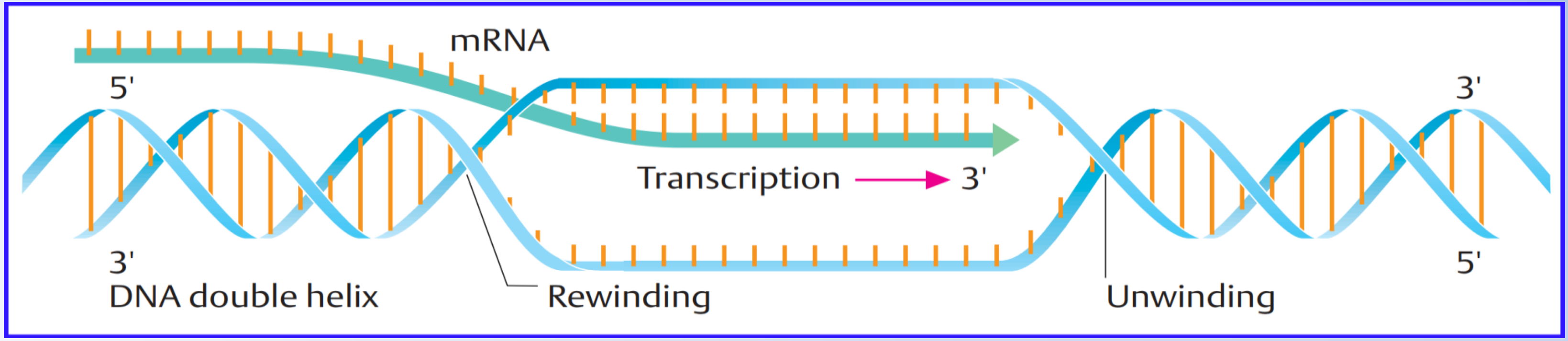
RAN Transcription

- The information of the coding sequences of a gene is transcribed into an intermediary RNA molecule, which is synthesized in sequences that are precisely complementary to the coding strand of DNA.

- Transcription of a particular gene always proceeds from one of the two DNA strands that acts as a template, the so-called (*Antisense)* strand and the RNA product is complementary to that template and called the *(Sense)* strand.

- The RNA transcription resulting a single-stranded RNA molecule acts as a mobile molecular copy of the original DNA sequence.

- The only difference is that in RNA, all of the T nucleotides are replaced with U nucleotides; during RNA synthesis, U is incorporated when there is an A in the complementary antisense strand.



Molecular Biology :

RNA Transcription: Stages of RNA Transcription:

Initiation:

- Transcription machinery binds to DNA and initiates transcription at initiation site .

- Transcription in prokaryotes and in eukaryotes requires the DNA double helix to partially unwind and forming the transcription bubble in the region of RNA synthesis.

- The DNA helix is opened by a complex set of proteins. The DNA strand in the 3! to 5! Direction (coding strand) serves as the template for the transcription into RNA, which is synthesized in the 5! to 3! Direction called the RNA sense strand .

- Like DNA polymerase, RNA polymerase adds nucleotides one by one to the *(3’-OH)* group of the growing nucleotide chain.

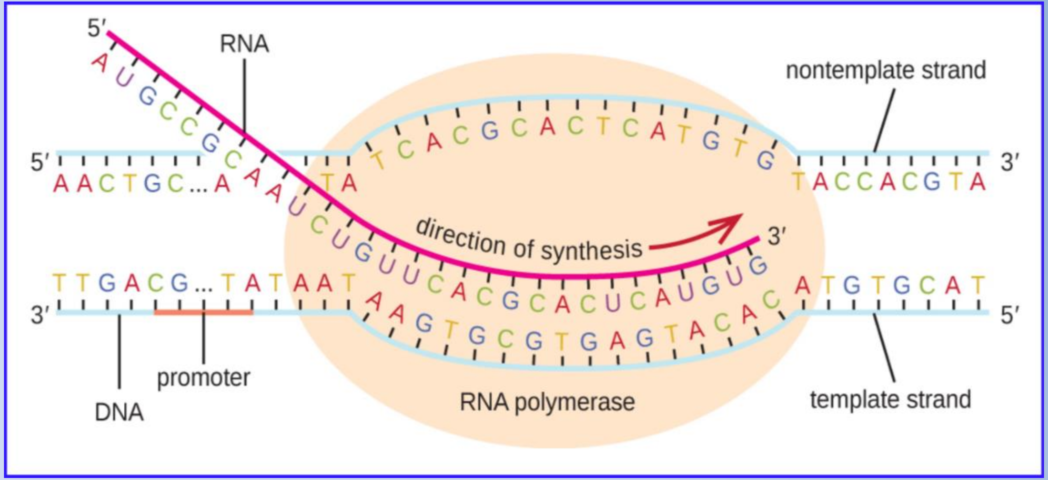
- One critical difference in between DNA and RNA polymerase is the DNA polymerase requires a 3’-OH group, whereas RNA polymerase does not to add nucleotides.

Molecular Biology :

RNA Transcription: Stages of RNA Transcription:

Elongation:

- The elongation in transcription phase begins when the *Sigma (σ) Subunit* dissociates from the polymerase, allowing the core enzyme to synthesize RNA complementary to the DNA template in a 5’ to 3’ direction at a rate of approximately 40 nucleotides per second. As elongation proceeds.
- DNA is continuously unwound ahead of the core enzyme and rewound behind it.
- Sigma (σ) subunit of bacterial RNA polymerase confers specificity as to which promoters should be transcribed.



Termination:

- Once a gene is transcribed, the polymerase must dissociate from the DNA template and liberate the newly made RNA. This is referred to as termination of *Transcription*.
- The DNA template includes repeated nucleotide sequences that act as termination signals, causing RNA polymerase to stall and release from the DNA template, freeing the RNA transcript.

Molecular Biology : *Fundamentals*

Gene expression: Protein Synthesis - Translation

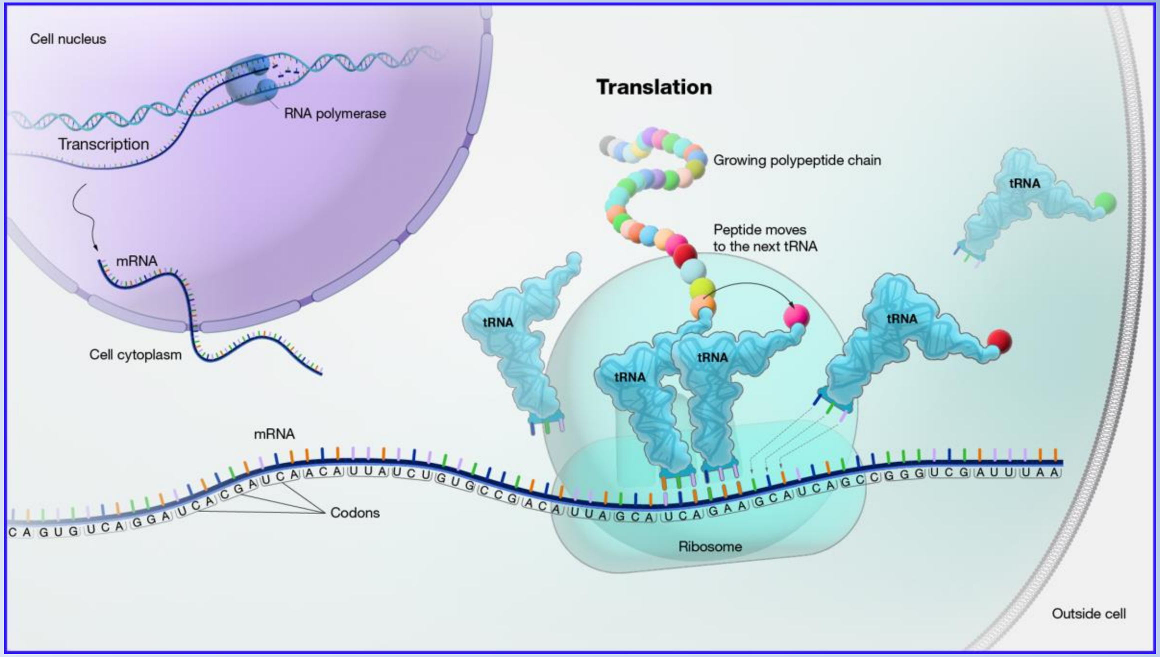
- During translation the sequence of codons made up of the nucleotide bases in mRNA is converted into a corresponding sequence of amino acids.

- The rRNA ensures the proper alignment of the mRNA, tRNA, in the ribosomes.

- The rRNA of the ribosome also has an enzymatic activity (peptidyl transferase) and catalyzes the formation of the peptide bonds between two aligned amino acids during protein synthesis.

Stages of Translation:

(1) Initiation:
an initiation complex comprising mRNA, a ribosome and tRNA is combination formed.



Gene expression: Protein Synthesis - Translation

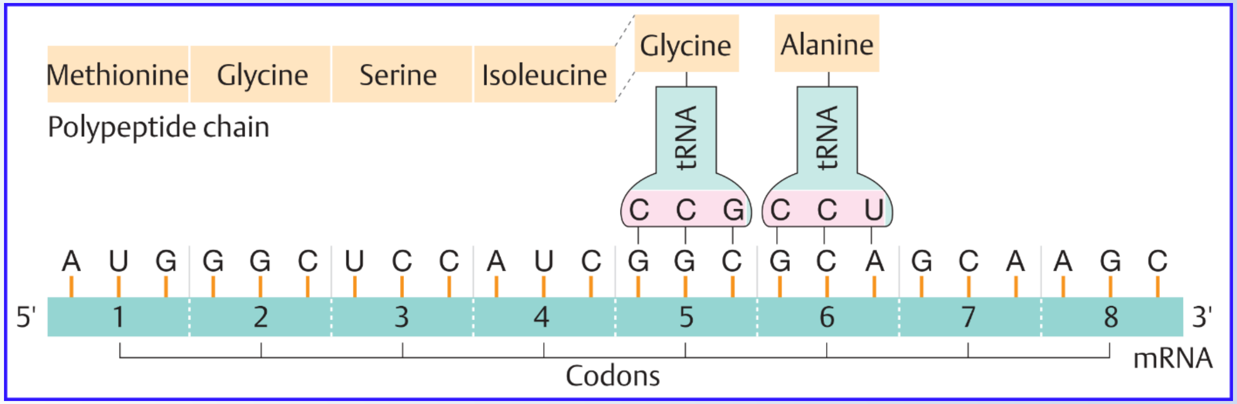
(2) <u>Elongation</u>:

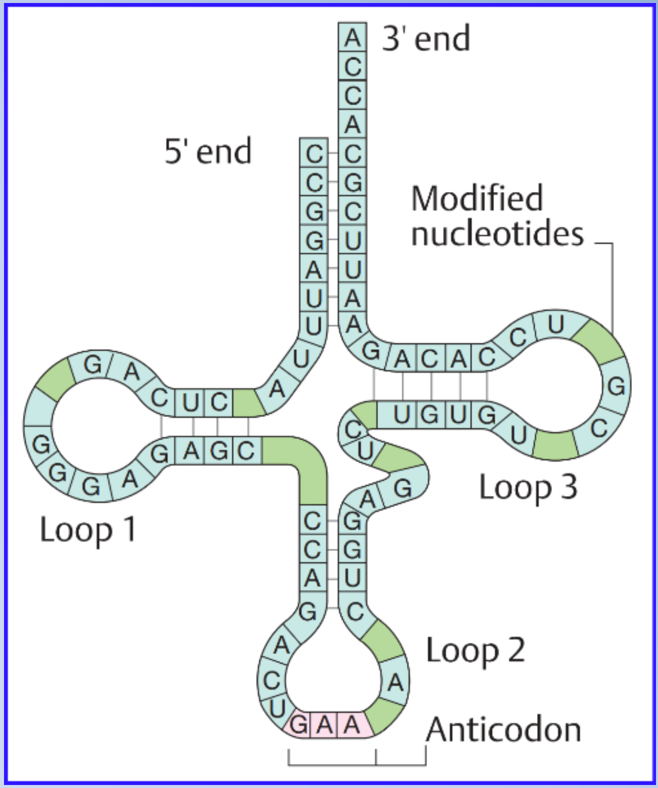
- Translation occurs in a reading frame which is defined at the start of translation (start codon **AUG**) and a further amino acid, determined by the next codon, is attached.

- It is the base pairing between the tRNA and mRNA that allows for the correct amino acid to be inserted in the polypeptide chain being synthesized.

- Each amino acid has its own tRNA

- Amino acids are joined in the sequence determined by the mRNA nucleotide bases and tRNA carries the correct amino acid to the site of protein synthesis in the ribosome.





Molecular Biology : Fundamentals

Gene expression: Protein Synthesis - Translation

(2) Elongation:

- Each amino acid has its own tRNA, which has a region that is complementary to its codon of the mRNA (anticodon).
- Movement (translocation) of the ribosome three nucleotides further in the 5'-3' direction of the mRNA.

(3) Termination:

when one of three mRNA stop codons (UAA, UGA, or UAG) is reached. The polypeptide chain formed leaves the ribosome.